DSA Experiment 10:Implement UART communication using Data Structures in C

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Batch:B1 **Class:SY1**

```
#include<stdio.h>
#include<stdint.h>
#include<stdbool.h>
#define UART_BUFFER_SIZE 64

typedef struct{
        uint8_t buffer[ UART_BUFFER_SIZE];
        int head;
        int tail;
        int count;
}UART_Buffer;

typedef struct{
        UART_Buffer txBuffer;
        UART_Buffer rxBuffer;

}UART_Device;

void UART_Buffer_Init(UART_Buffer *buf) {
   buf->head = 0;
   buf->tail = 0;
   buf->count = 0;
}
bool UART_Buffer_IsFull(UART_Buffer *buf) {
   return buf->count == UART_BUFFER_SIZE;
}

bool UART_Buffer_IsEmpty(UART_Buffer *buf) {
   return buf->count == 0;
}

bool UART_Buffer_Enqueue(UART_Buffer *buf, uint8_t data) {
   if (UART_Buffer_IsFull(buf)) return false;
   buf->buffer[buf->head] = data;
 buf->head = (buf->head + 1) % UART_BUFFER_SIZE;
   buf->count++;
   return true;
}
```

```
bool UART_Buffer_Dequeue(UART_Buffer *buf, uint8_t *data) {
    if (UART_Buffer_IsEmpty(buf)) return false;
    *data = buf->buffer[buf->tail];
    buf->tail = (buf->tail + 1) % UART_BUFFER_SIZE;
    buf->count--;
    return true;
}
void UART_Init(UART_Device *uart) {
    UART_Buffer_Init(&uart->txBuffer);
    UART_Buffer_Init(&uart->rxBuffer);
    printf("UART Initialized.\n");
}


void UART_SendByte(UART_Device *uart, uint8_t data) {
    if (UART_Buffer_Enqueue(&uart->txBuffer, data))
        printf("TX: '%c' enqueued\n", data);
    else
        printf("TX Buffer Full! Data lost.\n");
}


void UART_ReceiveByte(UART_Device *uart, uint8_t data) {
    if (UART_Buffer_Enqueue(&uart->rxBuffer, data))
        printf("RX: '%c' received\n", data);
    else
        printf("RX Buffer Full! Data lost.\n");
}


void UART_ProcessTransmit(UART_Device *uart) {
    uint8_t data;
    while (UART_Buffer_Dequeue(&uart->txBuffer, &data)) {
        printf("Transmitted: '%c'\n", data);
    }

}


void UART_ReadReceivedData(UART_Device *uart) {
    uint8_t data;
    while (UART_Buffer_Dequeue(&uart->rxBuffer, &data)) {
        printf("Read from RX: '%c'\n", data);
```

```
    }
}

int main() {
    UART_Device uart;
    UART_Init(&uart);

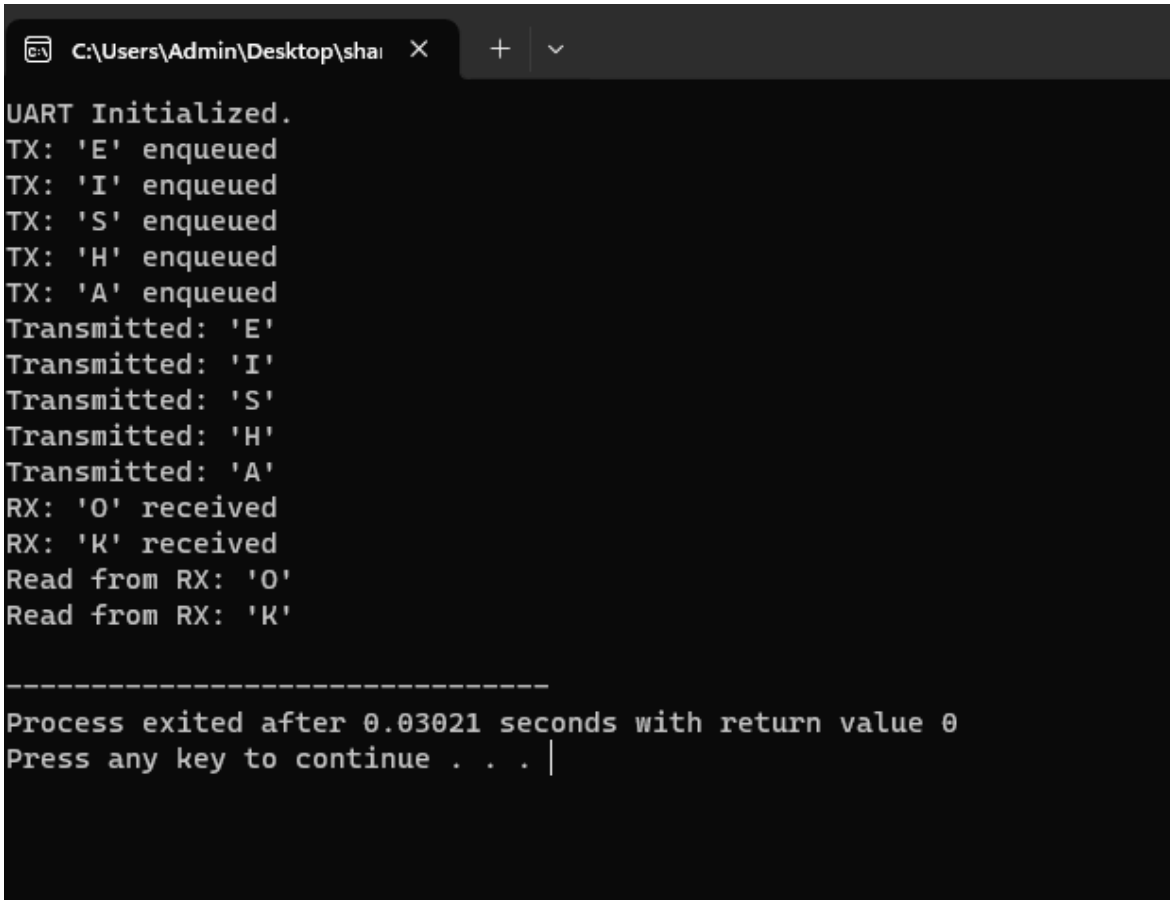

    UART_SendByte(&uart, 'H');
    UART_SendByte(&uart, 'E');
    UART_SendByte(&uart, 'L');
    UART_SendByte(&uart, 'L');
    UART_SendByte(&uart, 'O');


    UART_ProcessTransmit(&uart);


    UART_ReceiveByte(&uart, 'O');
    UART_ReceiveByte(&uart, 'K');


    UART_ReadReceivedData(&uart);

    return 0;
}
```


C:\Users\Admin\Desktop\sha
UART Initialized.
TX: 'E' enqueued
TX: 'I' enqueued
TX: 'S' enqueued
TX: 'H' enqueued
TX: 'A' enqueued
Transmitted: 'E'
Transmitted: 'I'
Transmitted: 'S'
Transmitted: 'H'
Transmitted: 'A'
RX: 'O' received
RX: 'K' received
Read from RX: 'O'
Read from RX: 'K'

Process exited after 0.03021 seconds with return value 0
Press any key to continue . . .